Lecture 5 - The Derivative

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Homework Updates/Info/Yappery?

After reading the engagement portfolios, I just had a few things I thought might be useful to say:

1. For choosing problems - I'd recommend looking over each of the problem types and choosing *a few problems from many problem types*.
2. Come review time (for quizzes and exams) the other problems you did not solve will be very useful practice problems (homework's will be uploaded on Canvas since evil gradescope doesn't let you see them after the due date)
3. If you're choosing homework problems in a way that's making you feel lost or making you feel unprepared for the quizzes please talk to me so we can discuss different strategies :)

Quiz Time!

Let’s get into nice orderly rows, spread out as much as possible, and slay this quiz :)

The Road So Far

So far we've developed this notion of a limit:

1. Limits $\lim_{x \to a} f(x)$ are *the values that $f(x)$ approximates as x approximates a to closer and closer degrees*
2. Functions whose *actual* values match their approximated values at $x = a$ (e.g. where $f(a) = \lim_{x \to a} f(x)$) are called *continuous at $x = a$*. c.f. continuous on $[a, b]$, and continuous
3. We developed *maaaany* different tools to compute limits.

Constant Value Approximation

For a function $f(x)$ that is *continuous at* $x = a$, how good is the approximation $f(x) \approx f(a)$ when $a \approx x$?

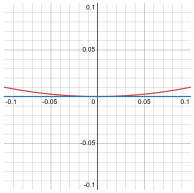


Figure: The graph of $f(x) = x^2$ near $x = 0$

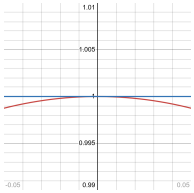


Figure: The graph of $f(x) = \cos(x)$ near $x = 0$

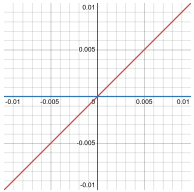


Figure: The graph of $f(x) = x$ near $x = 0$

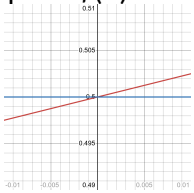


Figure: The graph of a logistic function near $x = 0$

What to do?

So if constant functions $y = f(a) \approx f(x)$ don't approximate well enough what's the next most complicated type of function we could approximate $f(x)$ by?
A linear function!

The Derivative

Linear Approximation

We will introduce the (extremely *extremely* important) concept of the derivative via *linear approximations*.

Goal

Find the linear function $\mathscr{L}(x) = mx + b$ that "*best*" approximates the function $f(x)$ when $x \approx a$ for some fixed value a.

Why? Oftentimes for scientific/engineering purposes, the actual functions that model complex phenomena (like pendulum forces, population growth rates, reaction rates, etc) are so wildly complicated that we cannot effectively study their true form.
However, taking their linear approximation (since linear functions are exactly specified by a finite amount of data [the slope and the intercept, e.g.]) often both simplify calculations vastly *AND* approximate real world measurements well enough.

Best Linear Approximation I

(Aside: Because we want to look at behavior when $x \approx a$, I'll switch to *point-slope form* $\mathscr{L}(x) = m(x-a)+b$)

Idea 1: We'll say that $\mathscr{L}(x) = m(x-a)+b$ is the best linear approximation of $f(x)$ near $x=a$ if $\lim_{x\to a} \mathscr{L}(x) = \lim_{x\to a} f(x)$. Pros/Cons? This at least forces $b = \lim_{x\to a} f(x)$.

Idea 2: ... if the error term $|f(x) - \mathscr{L}(x)| \to 0$ as $x \to a$. But this is just the last idea dressed in a different hat...

We can't *just* make conditions on limits of errors since the information about the linear term $m(x-a)$ "*dies*" as $x \to 0$.

Best Linear Approximation II

Solution: The error $|f(x) - \mathscr{L}(x)|$ should "die" to 0 faster than $|x - a|$ "dies" to 0.

Best Linear Approximation

A linear function $\mathscr{L}(x) = m(x - a) + b$ is the *best linear approximation* to a function $f(x)$ at $x = a$ if:

$$\lim_{x \to a} \frac{|f(x) - \mathscr{L}(x)|}{|x - a|} = 0 \tag{1}$$

That's great and all - but how do we *find* the best linear approximation?

Best Linear Approximation III

Determining the Best Linear Approximation

Given a function $f(x)$, if a best linear approximation $\mathscr{L}(x) = m(x-a)+b$ exists to $f(x)$ at $x=a$ then the following must be true:

1. $b = \lim_{x \to a} f(x)$
2. $f(x)$ is continuous at $x=a$, so
$$b = \lim_{x \to a} f(x) = f(a) \tag{2}$$
3. We have the following expression for the slope m:
$$m = \lim_{x \to a} \frac{f(x)-f(a)}{x-a} = \lim_{h \to 0} \frac{f(a+h)-f(a)}{h} = IV_{x=a}(f(x)) \tag{3}$$

Because this quantity has shown up sooooo many times, we give it a special name:

Derivative of a function at a point

The derivative of a function $f(x)$ at a point $x = a$, denoted $f'(a)$ is the common value of the following four (equal) quantities:

1. $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{4}$$
2. The slope of the best linear approximation to $f(x)$ at $x = a$.
3. The slope of the *tangent* line to the graph of $y = f(x)$ at $x = a$.
4. The instantaneous velocity of a particle with position described by $f(x)$ at time $x = a$.

Computing The Derivative

We often use Equation 4 to compute the derivative, which is why we spent the first two weeks going over limits in such depth!
E.g. let's compute the derivative of the function $f(x) = x^2$ at $x = 2$!

$$f'(2) = \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} \tag{5}$$

$$= \lim_{x \to 2} \frac{x^2 - 2^2}{x - 2} \tag{6}$$

$$= \lim_{x \to 2} \frac{(x-2)(x+2)}{x - 2} \tag{7}$$

$$= \lim_{x \to 2} x + 2 = 2 + 2 = 4 \tag{8}$$

$$f'(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} \tag{9}$$

$$= \lim_{h \to 0} \frac{(2+h)^2 - 2^2}{h} \tag{10}$$

$$= \lim_{h \to 0} \frac{4 + 4h + h^2 - 4}{h} \tag{11}$$

$$= \lim_{h \to 0} \frac{4h + h^2}{h} \tag{12}$$

$$= \lim_{h \to 0} 4 + h = 4 + 0 = 4 \tag{13}$$

Interpreting the Derivative

Does this actually give the slope of the best linear approximation/tangent line?

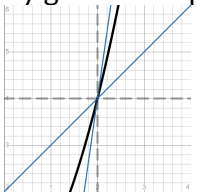


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 1, 7$

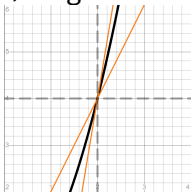


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 2, 6$

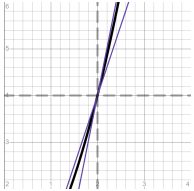


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 3, 5$

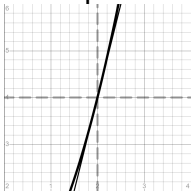


Figure: The best linear approximation of $f(x) = x^2$ around $x = 2$ with slope $m = f'(2) = 4$.

Computing the Derivative II

Let's try another one! Let's compute $f'(4)$ when $f(x) = \sqrt{x}$.

$$f'(4) = \lim_{x \to 4} \frac{f(x) - f(4)}{x - 4} \tag{14}$$

$$= \lim_{x \to 4} \frac{\sqrt{x} - \sqrt{4}}{x - 4} \tag{15}$$

$$= \lim_{x \to 4} \frac{\sqrt{x} + \sqrt{4}}{\sqrt{x} + \sqrt{4}} \frac{\sqrt{x} - \sqrt{4}}{x - 4} \tag{16}$$

$$= \lim_{x \to 4} \frac{(\sqrt{x})^2 - (\sqrt{4})^2}{(x - 4)(\sqrt{x} + \sqrt{4})} \tag{17}$$

$$= \lim_{x \to 4} \frac{x - 4}{(x - 4)(\sqrt{x} + \sqrt{4})} \tag{18}$$

$$= \lim_{x \to 4} \frac{1}{\sqrt{x} + \sqrt{4}} = \frac{1}{\sqrt{4} + \sqrt{4}} = \frac{1}{4} \tag{19}$$

$$f'(4) = \lim_{h \to 0} \frac{f(4 + h) - f(4)}{h} \tag{20}$$

$$= \lim_{h \to 0} \frac{\sqrt{4 + h} - \sqrt{4}}{h} \tag{21}$$

$$= \lim_{h \to 0} \frac{(4 + h) - 4}{(\sqrt{4 + h} + \sqrt{4})h} \tag{22}$$

$$= \lim_{h \to 0} \frac{h}{(\sqrt{4 + h} + \sqrt{4})h} \tag{23}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{4 + h} + \sqrt{4}} = \frac{1}{4} \tag{24}$$

Computing the Derivative III

For one last one, let's compute $f'(0)$ when $f(x) = |x|$.

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \tag{25}$$

$$= \lim_{x \to 0} \frac{|x|}{x} \tag{26}$$

since $|x|$ is defined piecewise, let's look at one-sided limits: (27)

$$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} \tag{28}$$

$$= \lim_{x \to 0^+} 1 = 1 \tag{29}$$

$$\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} \tag{30}$$

$$= \lim_{x \to 0^-} -1 = -1 \tag{31}$$

Since $\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0}$ this means that $f'(0)$ *does not exist* and there is ***NO*** best linear approximation to $|x|$ around $x = 0$.

Differentiable Functions I

Differentiable Function

A function $f(x)$ is *differentiable* at $x = a$ if any of the following equivalent conditions hold:

1. The limit
$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{32}$$
exists.
2. There exists a best linear approximation $\mathscr{L}(x) = f(a) + f'(a)(x - a)$ to $f(x)$ around $x = a$.
3. There is a unique tangent line to the graph of $y = f(x)$ at $x = a$.

Differentiable Functions II

The two classes of functions we've defined so far (continuous & differentiable) are related in the following manner:

Continuity and Differentiability

For a function $f(x)$ and a point $x = a$ we have the following (equivalent) relationships:

1. If $f(x)$ is differentiable at $x = a$ then it *must* be the case that $f(x)$ is also continuous at $x = a$.
2. If $f(x)$ is *not* continuous at $x = a$ then it *can never* also be the case that $f(x)$ is also differentiable at $x = a$.

Intuition for Continuity/Differentiability

The following intuition may be useful:

Intuition for Continuity and Differentiability

We can intuitively recognize continuity/differentiability of a function $f(x)$ at a point $x = a$ from its graph as:

1. continuity: if you can draw the graph of $f(x)$ near $x = a$ *without* lifting up your pencil (no jump discontinuities) or without rapidly oscillating it (no $\sin(\frac{1}{x})$)
2. differentiability: if the graph of $f(x)$ "locally looks like" the graph of a linear function near $x = a$ i.e. the function is both continuous and does not make any sharp changes near $x = a$.

Terminological Potpourri I

There's a lot of terminology so here's some scattered parts that didn't fit nicely.

Tangent Line

Given a function $f(x)$ and a point $x = a$, the *tangent line* to $f(x)$ at $x = a$ is the line that:

1. intersects the graph of $y = f(x)$ at $(a, f(a))$
2. but does not *cross* the graph at that point (e.g. it just "glances" off).

The tangent line *is* the graph of the best linear approximation to $f(x)$ at $x = a$, as such the equation of the tangent line, for $f(x)$ differentiable at $x = a$, is:

$$y = f(a) + f'(a)(x - a). \tag{33}$$

Terminological Potpourri II

Units of the Derivative

If our function represents the relationship between an input quantity x measured in units of U_{in} and an output quantity $f(x)$ measured in units of U_{out} the derivative $f'(a)$ (if it exists) is measured in units of $\frac{U_{out}}{U_{in}}$.

E.g. if $s(t)$ is the distance in miles that a car travels, where t is the time since starting measured in hours, $s'(t) = v(t)$ is the velocity and is measured in units of miles per hour.

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks!